

What is Penetration Testing as a Service (PTaaS)?

[Penetration Testing](#) as a Service (PTaaS) is a delivery platform, not to be confused with cloud pentesting. It offers more frequent and cost-effective access to penetration tests and a platform that facilitates collaborations between penetration testing service providers and client organizations. Organizations leverage PTaaS to detect and remediate vulnerabilities regularly.

In the past, penetration testing was a contract-based, complex effort that organizations could not carry out more than 1-2 times per year. PTaaS enables organizations to perform a penetration test daily, or even after each code change. While cloud penetration testing helps organizations identify security gaps in a certain cloud environment, PTaaS facilitates more frequent testing across all environments.

How PTaaS Works

In the past, penetration test results were delivered only after the testing period concluded. This information has merit, but the historical nature of the data can make it difficult for organizations to prioritize and fix according to the test results. The Software as a Service (SaaS) delivery model helps fix this issue, enabling organizations to run automated tests and view data on demand.

PTaaS vendors provide dashboards that include all the relevant data before, during, and after the test. Like traditional penetration testing services, PTaaS vendors offer resources for parsing vulnerabilities and verifying a remediation's effectiveness. Most PTaaS vendors provide a knowledge base to support in-house security teams in their remediations. Some vendors also offer assistance from the testers who discovered the vulnerability.

Organizations of all sizes can leverage PTaaS. Most PTaaS platforms can accommodate all business needs, including a full testing program and custom reporting features for regulatory compliance.

Learn more in our detailed guide to [pentesting tools](#)

Benefits of Pentesting as a Service

PTaaS aligns with modern software development methodologies, like DevOps, that require speed and agility from development and operations teams as well as security. Here are the notable benefits of PTaaS:

Hacker-Like Testing On Demand

A penetration test involves exploiting vulnerabilities by mimicking attackers. It enables organizations to learn how a threat actor perceives their current security posture and how existing security measures handle a real-life cyber attack. PTaaS initiates tests on demand, displaying the detected vulnerabilities as they are found and posted by the pentesters.

Early Feedback on Code Changes

The PTaaS model integrates seamlessly into the software development lifecycle (SDLC), providing developers with a vulnerability alert before they push new code to live environments. It enables teams to stay ahead of threat actors.

Fast Remediation Support

PTaaS platforms provide detailed remediation support, such as screenshots and videos, to assist organizations in locating and fixing vulnerabilities. This support saves significant time, eliminating the need to determine the issue and why it occurred.

Access to Security Engineers

PTaaS vendors can connect client organizations with security engineers to help fix security gaps. Getting access to this expertise helps ensure vulnerabilities get fixed without draining the resources of in-house teams.

Challenges of Using PTaaS

Here are the most common challenges of PTaaS:

Third-party restrictions

Not all third-party vendors provide pentesting continuously. Instead, they require their client organizations to request tests in advance. For example, Amazon Web Services (AWS) demands that customers obtain testing authorization in advance, allowing a maximum

window of 12 weeks. As a result, organizations can perform PTaaS in AWS regularly only if they ask for permission 4-5 times per year.

Sensitive data retention and handling

Each vendor handles sensitive data differently, but most use encryption to secure it. Since encryption processes typically use key management, it adds complexity for PTaaS vendors. As a result, the vendor might not be able to use the keys to archive data at rest.

Budget limitations

Automated orchestration enables organizations to manage internal resources and budgets efficiently, ensuring they can run more tests. However, underfunded and new security programs that struggle to remediate the vulnerabilities identified during annual penetration testing cannot handle shorter cycles.

What To Look For In a PTaaS Provider

Human, Hands-on Approach

Automated, software-driven solutions cannot find all critical vulnerabilities in an environment or a software application. Human expertise offers more flexibility and creativity to support manual testing in rooting out sophisticated vulnerabilities and cyber attacks that automation might miss. Human intelligence can instinctively sense when to investigate deeper and when to move on. PTaaS vendors that offer manual testing can cover more ground, offering more comprehensive coverage.

Dedicated Expertise

A penetration testing service relies on the experts conducting these tests. The ideal PTaaS vendor hires talent with experience and qualifications to support organizations. Certifications like OSCP, OSCE, and OSWE can help assess the qualifications of the vendor's experts.

Some PTaaS vendors rely on a crowdsourced model that assigns a different penetration tester to the organization every time. As a result, organizations cannot form a consistent relationship with a tester who thoroughly understands the organization's estate and applications.

Additionally, a crowdsourced model reduces standardization, which means testers cannot perform the same actions repeatedly to optimize results and provide faster outcomes. However, it does diversify testing to uncover vulnerabilities that the same tester may miss year after year.

Useful, Actionable Reporting

A penetration test should provide reporting capabilities that stakeholders can understand and act on. A report must provide a high-level executive summary and a more detailed technical view of all findings, covering impact, risks, vulnerability details, proofs of concept, attack vectors, mitigation recommendations, and prioritized remediation paths.

DevSecOps Friendly

PTaaS helps support DevSecOps teams when shifting security left. Testing applications at an early phase and testing repeatedly enables teams to solve security problems as they occur. As a result, DevSecOps teams can create a more secure application without going through costly rebuilding during late SDLC phases.

PTaaS vendors offer dashboards that display information for technology, security, and business teams. It provides the information needed to reduce vulnerability remediation lead time and increase visibility into potential risks. Dashboards can save direct costs, offering best-in-class features, controls, and configurations. The ideal dashboard seamlessly integrates with existing clouds and technology stacks.

How HackerOne Can Help

HackerOne's attack resistance management security helps keep businesses safe. HackerOne Pentest helps organizations to go beyond the status quo of "check the box" penetration testing. Organizations using HackerOne Pentest benefit from fast, actionable security outcomes with on-demand pentest-as-a-service (PTaaS) capabilities. Pentest engagements launch in as few as seven to ten days, compared to three to four weeks for traditional pentests.

HackerOne customers are able to find the critical vulnerabilities that others miss by leveraging pentesters curated from our talented community of ethical hackers. Pentesters for HackerOne have 3+ years of experience and are OSCP, OSE, and OSWE certified. 65% have 5+ years of experience. HackerOne Pentest is even more effective at keeping businesses safe when combined with the full attack resistance management portfolio.

[Learn more about HackerOne PTaaS](#)

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